Model Evaluation

Nested Models

Last class:

Goodness of fit

- RSS, ESS, TSS

- the coefficient of determination, R^2

- we introduced a new case study from Biology aiming to study the relation between mRNA and protein levels

- we evaluated the goodness of fit of SLR and MLR for the data and found a poor fit of the proposed model for most genes

Evaluation metrics

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Many metrics used to evaluate LR measure how far y is from $\hat{y}$:

- **Residuals Sum of Squares**: RSS = $\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$

> sum of the squared residuals, small is good

- **Residuals Standard Error**: RSE = $\sqrt{\frac{1}{n-p-1}\mathrm{RSS}}$

> gives an idea of the size of the *irreducible* error, very similar to the RSS, small is good

- **Mean Squared Error**: MSE = $\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$

> mean of the squared residuals, small is good. Usually computed on both the *training* and the *test* sets.

Coefficient of Determination

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All the measures listed above are *absolute* measures

It is not easy to judge if these are small enough. Important relative measures:

- **Coefficient of Determination,** $R^2 = 1 - \frac{RSS}{TSS}$

> it also compares y vs $\hat{y}$ using the RSS but relative to the TSS (sum of the squared residuals of the intercept-only model)

> it is a common measure of goodness of fit: how well does the model fit the *training* data?

> it doesn't have a known distribution so you can't use it to test a statistical hypothesis

> it increases with the size of the model so it is not useful to compare models of different sizes

Adjusted R^2

$$\text{adj}R^2 = 1 - \frac{RSS/(n-p-1)}{TSS/(n-1)}$$

> it is a common measure of goodness of fit: how well does the model fit the *training* data?

> penalizes the RSS by the size of the model so it can be used to compare models of different sizes

Today

Comparing nested models

- F-test for goodness of fit

- a more general F-test

The *F*-test

Although the R^2 is a widely used measure to evalute goodness of fit, it is not a formal statistical test.

Test: is the full model significantly different from an intercept-only model?

> The intercept-only model does not include any covariate, thus the estimated coefficient is given by the mean of the response, $\bar{Y}$, in our case the average protein value

The test hypotheses

- model reduced: $Y_i = \beta_0 + \varepsilon_i$
- model full: $Y_i = \beta_0 + \beta_1 X_{i1} + \ldots + \beta_p X_{ip} + \varepsilon_i$

We are *simulataneously* testing if many coefficients are zero!!

$$H_0 : \beta_1 = \beta_2 = \ldots = \beta_p = 0$$

against

$$H_1 : \text{at least one } \beta_j \neq 0 \text{ (for } j = 1, 2, \ldots, p)$$

The F-statistic

$$\frac{(RSS_{reduced} - RSS_{full})/p}{RSS_{full}/(n-p-1)} \sim \mathcal{F}_{p,n-p-1}$$

- $RSS_{reduced}$ is the **RSS** of the reduced model
- RSS_{full} is the **RSS** of the full model
- p is the number of covariates in the full model (excluding the intercept), number of parameters tested

Don't worry, R computes this statistic for you!!

The F-test in R

There are 2 functions in R that we can use to perfor this test:

- `glance(model_full)`

- `anova(model_reduced, model_full)`

> With `glance()` the reduced model is *always* the intercept-only model

> With `anova()` you can compare the full against any reduced model

Back to the case study

The test for model.2

- model reduced: $\text{prot}_i = \beta_0 + \varepsilon_i$

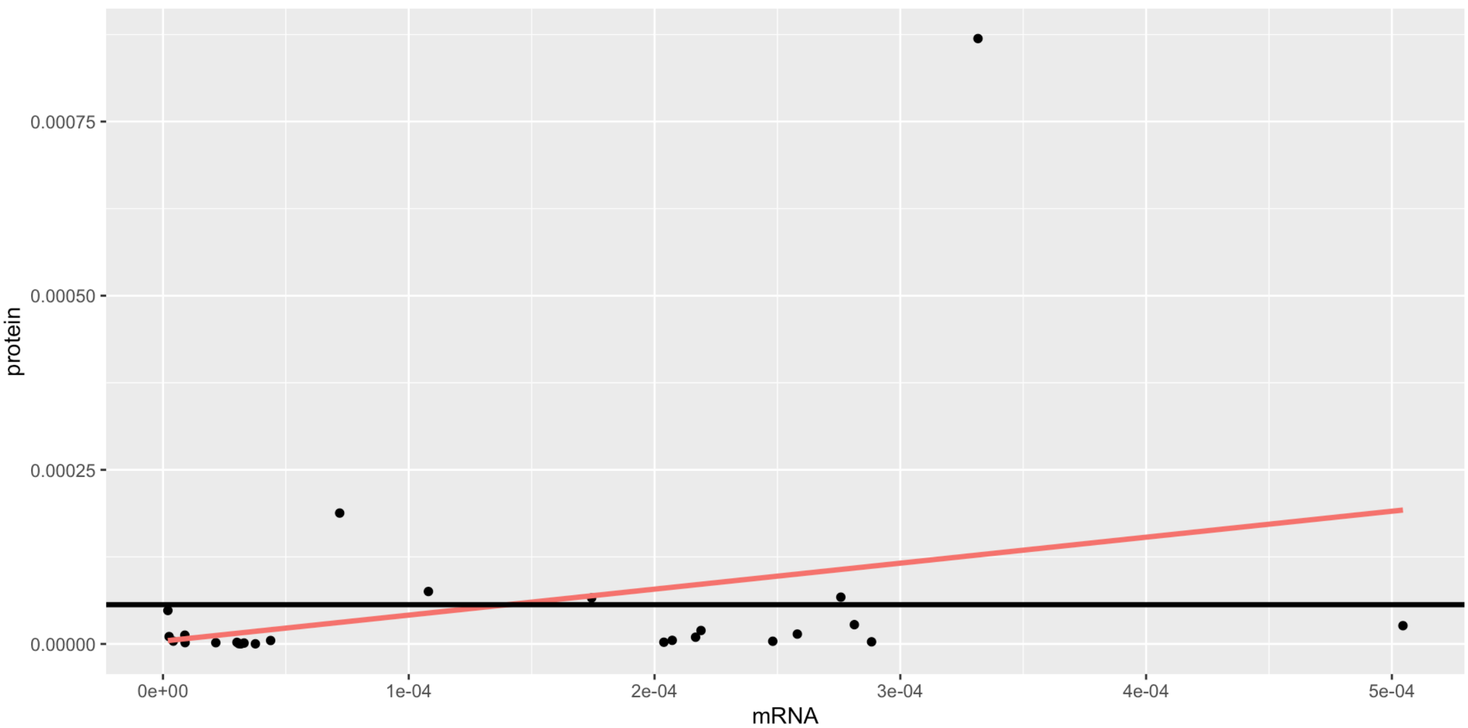
- model.2 full: $\text{prot}_i = \beta_0 + \beta_1 \text{mrna}_i + \varepsilon_i$

We are *simulataneously* testing if many coefficients are zero!!

$$H_0 : \beta_1 = 0$$

$$H_1 : \beta_1 \neq 0$$

> Note that these are the same hypotheses tested by `lm`!!


0.00075
0.00050
0.00025
0.00000
protein
0e+00
1e-04
2e-04
3e-04
4e-04
5e-04
mRNA

ANOVA

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
25	0	NA	NA	NA	NA
24	0	1	0	2.285	0.144

GLANCE

r.squared	adj.r.squared	sigma	statistic	p.value	df	logLik	AIC	BIC	deviance	df.residual
0.087	0.049	0	2.285	0.144	1	190.382	-374.764	-370.99	0	24

lm-tidy

term	estimate	std.error	statistic	p.value
(Intercept)	0.000	0.000	0.086	0.932
mrna	0.373	0.247	1.512	0.144

t-test vs F-test

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t-test

`lm()` computes t-tests to evaluate the contribution of *each variables* to explain a response ***with possibly other variables included*** in the model!!

$$H_0 : \beta_j = 0, \text{ versus } H_1 : \beta_j \neq 0$$

> H_0 contains only *one* coefficient, even if the model has more

The results of the t-tests evaluate the contribution of *each* variable (separately) to explain the variation observed in the response

F-test

`glance()` computes an F-test to evaluate the contribution of *all variables* to explain a response (all *vs* nothing)!!

$$H_0 : \beta_1 = \beta_2 = \ldots = \beta_p = 0, \text{ versus } H_1 : \text{at least one } \beta_j \neq 0$$

> H_0 contains *all* coefficient (except the intercept)

Special case: $p = 1$

If there is only covariate in the full model, the hypotheses are the same!

> It can be shown that the F-test (by `glance`) is equivalent to the t-test (by `lm`): $F = t^2$ and equal p-values

> Both tests (F and t) are based on Normality assumptions or approximate Normality distribution given by large sample theory

The test for model.4

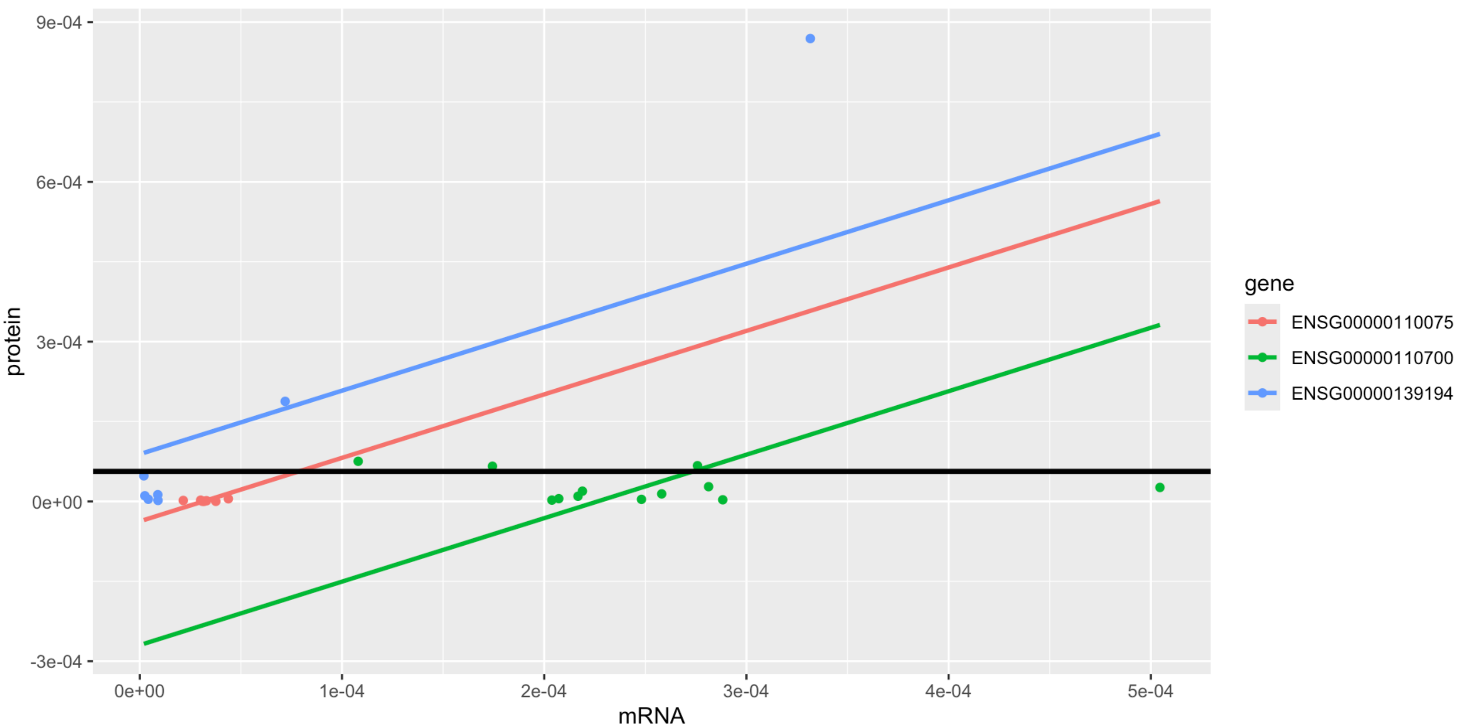
- model reduced: $\text{prot}_i = \beta_0 + \varepsilon_i$
- model.4 full: $\text{prot}_t = \beta_0 + \beta_1 \text{mrna}_t + \beta_2 \text{gene2}_t + \beta_3 \text{gene3}_t + \varepsilon_t$

We are *simulataneously* testing if many coefficients are zero!!

$$H_0 : \beta_1 = \beta_2 = \beta_3 = 0$$

$$H_1 : \text{at least one } \beta_j \neq 0, \text{ for } j = 1, 2, 3$$

> These are **not** the same hypotheses tested by `lm`!!



9e-04
6e-04
3e-04
0e+00
-3e-04
protein
0e+00
1e-04
2e-04
3e-04
4e-04
5e-04
mRNA
gene
ENSG00000110075
ENSG00000110700
ENSG00000139194

ANOVA

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
25	0	NA	NA	NA	NA
22	0	3	0	7.941	0.001

GLANCE

r.squared	adj.r.squared	sigma	statistic	p.value	df	logLik	AIC	BIC
0.52	0.454	0	7.941	0.001	3	198.738	-387.476	-381.186

lm-tidy

term	estimate	std.error	statistic	p.value
(Intercept)	0.000	0.00	-0.770	0.450
mrna	1.192	0.29	4.112	0.000
geneENSG00000110700	0.000	0.00	-2.681	0.014
geneENSG00000139194	0.000	0.00	1.859	0.076

Interpretation:

For these 3 genes, the p-value of the F-test is less than 0.05.

There is enough evidence to reject the null hypothesis.

The prediction from the additive model (common slope and different intercepts) is significantly better than the average protein value from the intercept-only model.

IMPORTANT: this doesn’t mean that we can predict protein from mRNA!

There is another variable in the model `gene` that may be as (or more) important as `mRNA`

More general

The F-test can be used to compare any pair of nested models

> LR with $q+1$ coefficients

- model reduced: $Y_i = \beta_0 + \beta_1 X_{i1} + \ldots + \beta_q X_{iq} + \varepsilon_i$

> LR with $p+1$ coefficients, k additional explanatory variables

- model full: $Y_i = \beta_0 + \beta_1 X_{i1} + \ldots + \beta_q X_{iq} + \ldots + \beta_p X_{ip} + \varepsilon_i$

Case example

For example:

- model.2 reduced: `lm(prot ~ mrna)`

$$\text{prot}_i = \beta_0 + \beta_1 \text{mrna}_t + \varepsilon_i$$

- model.4 full: `lm(prot ~ mrna + gene)`

$$\text{prot}_t = \beta_0 + \beta_1 \text{mrna}_t + \beta_2 \text{gene2}_t + \beta_3 \text{gene3}_t + \varepsilon_t$$

The test hypotheses

We are *simulataneously* testing if many parameters (all the additional ones) are zero!!

$$H_0 : \beta_{q+1} = \beta_{q+2} = \ldots = \beta_p = 0$$

against

$$H_1 : \text{at least one } \beta_j \neq 0 \text{ (for } j = q+1, q+2, \ldots, p\text{)}$$

The F-statistic

$$\frac{(RSS_{reduced} - RSS_{full})/k}{RSS_{full}/(n-p-1)} \sim \mathcal{F}_{k,n-p-1}$$

- $RSS_{reduced}$ is the **RSS** of the reduced model
- RSS_{full} is the **RSS** of the full model
- k is the number of parameters tested (#full - #reduced)
- p is the number of covariates in the full model (excluding the intercept)

Don't worry, R computes this statistic for you!!

The F-test in R

For any reduced model, other than the intercept-only:

- `anova(model_reduced, model_full)`

> With `glance()` the reduced model is *always* the intercept-only model

In our example

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- model.2 reduced: `lm(prot ~ mrna)`,
  $\text{prot}_i = \beta_0 + \beta_1 \text{mrna}_t + \varepsilon_i$

- model.4 full: `lm(prot ~ mrna + gene)`,
  $\text{prot}_t = \beta_0 + \beta_1 \text{mrna}_t + \beta_2 \text{gene2}_t + \beta_3 \text{gene3}_t + \varepsilon_t$

$$H_0 : \beta_2 = \beta_3 = 0$$

$$H_1 : \text{at least one } \beta_j \neq 0, \text{ for } j = 2, 3$$


```
model2_red <- lm(prot ~ mrna, dat_3genes)

model4_full <- lm(prot ~  mrna + gene, dat_3genes)

anova(model2_red, model4_full) %>%
  mutate_if(is.numeric, round, 3) %>%
  knitr::kable()
```

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
24	0	NA	NA	NA	NA
22	0	2	0	9.919	0.001

Model Selection

Do we need all the available predictors in the model?

Some datasets contain *many* variables but not all are relevant

> you may want to identify the *most relevant* variables to build a model

People use the F-test and even the t-tests to select significant variables and then re-fit the model using the selected variables

Warning

Caution: the data is used over (and over) again to select so inference results are not longer valid. Post-inference problem

Summary

- The R^2, coefficient of determination, can be used to compared the sum of squares of the residuals of the fitted model with that of the intercept-only model

- The R^2 is usually interpreted as the part of the variation in the response explained by the model

- Many definitions and interpretations of the R^2 are for LS estimators of LR containing an intercept

- The R^2 can not be used to compare models of different sizes since bigger models always have larger R^2.

- Instead, the adjR^2 can be computed to take into account the sizes of the models.

- The R^2 is not a test and it does not provide a probabilistic result and it's distribution is unknown!

- Instead, we can use an F test, using `anova()`, to compare nested models
  - tests the simultaneous significance of additional coefficients of the full model (not in the reduced model)
  - in particular, we can use it to test the significance of the full model over the intercept-only model (using `glance()` or `anova()`)

- These F tests can be used to select among nested models.

> ⚠ **Warning**
>
> Caution: you may need to partition the data to avoid a post-inference problem (more coming in week 10)